

AP

NUCLEAR MILESTONES

1956
1 MWt (megawatt thermal) **APSARA research reactor** attains criticality

1957
Atomic Energy Establishment, Trombay, set up; later renamed **Bhabha Atomic Research Centre** (BARC)

1960
40 MWt **CIRUS reactor** attains criticality and begins making weapons-grade plutonium

1962
Heavy water production begins at German-built Nangal plant

1963
The US and India sign an accord stipulating that the former will supply enriched fuel to the Tarapur nuclear power plant

1968
India refuses to join the Nuclear Non-Proliferation Treaty; **Nuclear Fuel Complex** set up in Hyderabad

1973
The Canadian-built 100 MWe (megawatt electrical) heavy water reactor **Rajasthan-1 begins operations** at Rajasthan Atomic Power Station (RAPS)

1974
India conducts an underground "peaceful nuclear explosion", code-named **Smiling Buddha**, at Pokhran, Rajasthan

1982
BARC's Power Reactor Fuel Reprocessing Plant at Tarapur is commissioned

1985
The heavy water 100 MWt Dhruva reactor attains criticality and starts producing weapons-grade plutonium

1997
PM I.K. Gujral says India will not sign the Fissile Material Cut-off Treaty (FMCT) or any other "discriminatory" nuclear agreement

1998
India conducts two rounds of nuclear weapon tests at Pokhran

1999
DRDO chief A.P.J. Abdul Kalam says "the Agni II (intermediate-range ballistic missile) is designed to carry a nuclear warhead if required" and claims that an Agni-class payload was tested during the nuclear test at Pokhran in 1998

2015
DBT launches NBDS-II, a five-year programme to turn India into a biotech hub

2017
Government approves the construction of 10 700 MW heavy water reactors

2018
The International Panel on Fissile Material estimates that India has a net stockpile of 0.58 MT of weapons-grade plutonium, enough for 100-150 nuclear warheads